Quiz: 1 B   Name:

Part A: File System Navigation

Consider the file system below where the root is `~`, your home directory. The shapes following the names aren't part of the file system; they're pointers used in the questions below.

```
~
└── fall ○
    ├── span120A △
    │   ├── syllabus.txt ▲
    │   ├── essay1.txt
    │   └── essay2.txt
    ├── math110
    │   ├── syllabus.docx ●
    │   └── problem-sets
    │       ├── ps1.txt
    │       └── ps2.txt
    └── stat133
        ├── syllabus.docx
        └── quizzes ▽
            ├── quiz1.qmd ■
            ├── quiz2.qmd
            └── readme.txt
```

1. What is the absolute path to the file ●?

2. What is the relative path from the directory ▽ to the file ■?

3. What is the relative path from ▽ to ▲?

4. If you are in a UNIX shell and your working directory is ○, how can you print the content of ▲ using only one line of code?

5. If you are in a UNIX shell and your working directory is △, how can you create a new directory named `summer` in your home directory `~` using only one line of code?

6. If you are in a UNIX shell and your working directory is ▽, how can you copy file ■ into your newly created `summer` folder using only one line of code?

Part B: File System in R (use the same tree above)

7. Starting from your home directory `~`, write R code to change the working directory to △.

8. After changing the working directory, write the R command to display your current working directory.

9. Assume your current working directory is ▽. Write R code to read the file ▲ line by line into an object called `final`.

Part C: Working with Vectors and Matrices

10. Write an R command that creates the following vector (without using `c()`):

```
[1]  2  4  6  8 10 12 14 16
```

11. Predict the output of this code:

```r
1:4 + 1:2
```

12. Predict the output of this code:

```r
c(4,5,6) ^ c(1,2)
```

13. Predict the output of this code:

```r
mean(c(T,T,FALSE,T,FALSE))
```

14. Using `z <- c("apple" = 2, "coconut" = 4, "cherry" = 10)`, write the subsetting command that returns:

```
cherry coconut  apple
    10       4      2
```

(note: there are multiple ways to do this)

15. Using `z <- c("apple" = 2, "coconut" = 4, "cherry" = 10)`, write the subsetting command that returns:

```
coconut coconut
      4       4
```

(note: there are multiple ways to do this)

16. What is the data type of the vector `TGIF <- c("movies", "chicken wings", 100L, NA , NULL)`?

17. Write the R command that creates a dimension `3x2` (rows by columns) matrix object named `mtx1`, using the vector `c(2,3,4,5,3,4)`, with the contents filled by column.

18. For the `mtx1` matrix you created in question 17, write the R command that reassign the element `5` to `2` (note: there are multiple ways to do this).

19. Predict the result (write it as a matrix) of the R command:

```
matrix(c(2,4,6,8),2,2)/2
```